

AegisBorder AI — Smart Document & Identity Screening Platform

Comprehensive Technical Architecture, Verification Engine & Operational Walkthrough

Repository: <https://github.com/Biswarup-das23/AI-Based-Fake-Identity-Document-Screening-System> • **Compliance:** ICAO Doc 9303 (TD1, TD2, TD3) • **Status:** Production Ready • **Author:** Biswarup Das

1. Executive Summary & Operational Background

Modern international border security checkpoints process hundreds of thousands of identity documents daily, including national passports, visas, transit permits, and identity credentials. Traditional verification relies heavily on visual manual inspection and basic optical readers, introducing critical vulnerabilities and substantial transit delays:

Operational Challenge	Vulnerability Vector	AegisBorder AI Automated Solution
Fake Passports & Visas	Counterfeit booklet stock, fabricated guilloché patterns, invalid check digits	ICAO 9303 mathematical check-digit auditing with 7-3-1 weighting algorithm.
Altered Photographs	Digitally spliced or physically replaced portrait stickers to evade watchlist alerts	Error Level Analysis (ELA) heatmaps and Sobel boundary edge discontinuity jump analysis.
Modified Dates of Birth	Printed visual zone date modified to conceal minor/senior identity or bypass age restrictions	Cross-field reconciliation engine matching visual OCR against encrypted MRZ checksums.
Identity Impersonation	Traveler presenting legitimate document of a look-alike sibling or stolen passport	1:1 Cosine facial biometric matching of document portrait against live terminal camera stream.
Blacklisted Fugitives	Individuals subject to Interpol Red Notices or travel bans using forged documents	Sub-second real-time indexing against simulated international intelligence watchlists.
High Volume Delays	Human inspection bottlenecks causing 3–5 minutes wait time per passenger	End-to-end multi-module pipeline execution in under 1.2 seconds per traveler.

2. System Architecture & Multi-Spectral Pipeline

AegisBorder AI employs a modular micro-service inspired architecture pairing a high-speed Python/FastAPI computational backend with a reactive React 19 / Tailwind CSS v4 glassmorphic border control interface. The screening engine executes four specialized modules synchronously in memory:

Screening Pipeline Stage	Underlying Technologies	Deliverables & Outputs
Stage 1: Dual Ingestion	Multi-spectral document scanner upload + Live camera stream (Webcam/Sensor)	Synchronous acquisition of document scan matrix and live traveler portrait.
Stage 2: Module 1 (OCR/MRZ)	Morphological region localization, TD1/TD2/TD3 ICAO 9303 parser, VIZ regex extractor	Extracted fields (Name, Doc #, DOB, Expiry, Sex, Nationality) + ICAO checksum audit.
Stage 3: Module 2 (Forensics)	Error Level Analysis (ELA Q=90), Sobel edge gradient, Laplacian noise variance, EXIF	Multi-spectral visual heatmaps (ELA, Noise Discrepancy, Edge Jump) + Tamper Scores.
Stage 4: Module 3 (Biometrics)	Haar cascade/skin geometry detection, 1:1 Cosine landmark matching, 2D FFT PAD	Biometric similarity score (0–100%), presentation attack / moiré detection.
Stage 5: Module 4 (Watchlist)	Exact & fuzzy hash matching against Interpol Red Notices and border exclusion databases	Watchlist flag status, severity categorization (CRITICAL / HIGH), detention mandates.

Stage 6: Risk & Decision Engine	Weighted Bayesian composite scoring engine + SHA-256 cryptographic audit ledger	Composite Risk Index (0–100%), decision tier, and printable cryptographic certificate.
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3. Technical Deep Dive: The 4 Core AI Modules

3.1 Module 1: ICAO Doc 9303 Check Digit Engine & Integrity Check

The International Civil Aviation Organization (ICAO) Doc 9303 standard defines strict mathematical rules for travel documents. All characters map to numerical values (0–9 = 0–9, A–Z = 10–35, '<' = 0). A cyclic 7-3-1 weight pattern is applied across each field string S to compute check digit C :

Formula: $C = (\sum_{i=0}^{n-1} Value(S_i) \times Weight_{i \bmod 3}) \bmod 10$ where $Weight \in [7, 3, 1]$

The engine audits document number check digit, birth date check digit, expiry date check digit, optional data check digit, and the composite check digit over Line 2. Discrepancies between the visual text (VIZ) and MRZ immediately flag tampering.

3.2 Module 2: Multi-Spectral Forensics (ELA, Edge Discontinuity, Noise)

Error Level Analysis (ELA): Re-compresses the document image at JPEG quality $Q=90$. Original undisturbed regions settle at uniform low-amplitude error rates, while spliced or modified pixels (such as pasted avatar photos or retouched text) exhibit bright high-frequency error spikes.

Portrait Boundary Step Discontinuity: Spliced portrait replacement creates unnatural sharp edges against the security guilloché background. The engine computes Sobel gradient magnitude $G = \sqrt{G_x^2 + G_y^2}$ along the four portrait boundaries. Discontinuity magnitude $G_{edge} > 65.0$ triggers immediate forgery alerts.

Laplacian Noise Variance: Computes local noise standard deviations across segmented patches to identify non-uniform sensor noise characteristics indicative of cloned or AI-generated digital inserts.

3.3 Module 3: 1:1 Facial Biometric Verification & Presentation Attack Detection

1:1 Face Matching: Crops the document avatar and live passenger camera frame, extracts normalized spatial geometry and gradient histograms, and computes Cosine vector similarity: $Sim(u, v) = (u \cdot v) / (||u|| ||v||)$. Similarity $\geq 70\%$ confirms positive identity; scores below 50% signal identity impersonation.

Anti-Spoofing (PAD): Electronic displays (smartphones, iPads) and printed card re-photos emit periodic pixel grid artifacts. A 2D Fast Fourier Transform (FFT) detects anomalous localized ring peaks in the high-frequency spectrum, defending against presentation attacks.

3.4 Module 4: Watchlist & Interpol Red Notice Screening

Indexes traveler name, document number, and nationality against an international law enforcement ledger. Matches flag high-severity red notices, active entry bans, or stolen document series, triggering automated detention mandates.

4. Real-World (IRL) Testing & 'New Passenger' Ingestion System

To facilitate real-world trials, live demonstrations, and field edge-case analysis, the system features a dedicated **New Passenger Registration Console** accessible directly from the dashboard header and preset bar:

- Dual-Channel Media Ingestion:** Operators can upload actual document scans/photos and capture live traveler selfies using a connected webcam or mobile camera feed.
- Real-Time Document Synthesis:** When physical scans are omitted, the system generates high-definition synthetic ICAO Doc 9303 compliant documents embedding the uploaded passenger portrait into security guilloché patterns.
- Automated ICAO Checksum Generator:** Generates mathematically valid TD3 (Passport) or TD2 (Visa) MRZ strings with composite check digits based on user-entered metadata.

- **Threat Scenario Injector:** Operators can test system edge-cases by selecting injected threat scenarios: *Authentic (Clean)*, *Photo Tampering (ELA trigger)*, *Checksum Forgery (Check digit error)*, *DOB Mismatch (VIZ vs MRZ conflict)*, or *Interpol Red Notice Hit*.
- **Dynamic In-Memory Session Ledger:** Newly registered IRL passengers appear immediately in the screening presets bar with an 'IRL PASSENGER' badge, enabling instant re-screening and deletion.

5. Unified Risk Scoring Matrix & Officer Protocols

The Composite Risk Score *R* (0–100%) integrates weighted evidence across all modules with security overrides:

Risk Formula: $R = 0.25 \times R_{integrity} + 0.35 \times R_{forensics} + 0.20 \times R_{biometrics} + 0.20 \times R_{watchlist}$
Automated Override Logic: If an Interpol Red Notice matches, risk overrides to ≥ 92%. If portrait tampering exceeds 75%, risk overrides to ≥ 85%. If critical checksums fail, risk floor is set to ≥ 62%.

Score Range	Risk Tier	Officer Action	Border Operational Protocol
0 – 25%	LOW RISK	GRANT ENTRY	All ICAO checksums pass, zero tampering, 1:1 face verified, clear watchlist. Issue automated electronic entry stamp.
26 – 55%	MODERATE	INTERVIEW / REVIEW	Minor compression noise or borderline face match. Officer requests secondary identity proof (boarding pass/national ID).
56 – 80%	HIGH RISK	SECONDARY INSPECT	Invalid ICAO checksum, expired credentials, or DOB mismatch. Route traveler to Secondary Inspection Counter 4B.
81 – 100%	CRITICAL	DETAIN SUBJECT	Confirmed photo replacement, Interpol Red Notice, or presentation attack. Immediate security detention & document confiscation.

6. Walkthrough of Test Scenarios & Validation Results

The screening platform was verified against 5 built-in border checkpoint threat presets and multiple custom IRL profiles:

Scenario Profile	Traveler Identity	Primary Anomaly Detected	Risk Score	Mandated Action
Preset 1: Genuine Passport	Sarah Helena Schmidt (DEU • C44L28901)	Zero anomalies. All checksums valid. 94.2% facial similarity. Clean record.	2.8% (LOW)	GRANT ENTRY
Preset 2: Tampered Photo	Alexander Cross (GBR • P90812455)	ELA artifact spike (78%), boundary edge discontinuity (88.5%), face mismatch (42%).	85.0% (CRITICAL)	DETAIN SUBJECT
Preset 3: Forged Visa	Tariq Mansur (EGY • V33918204)	Mathematical check digit failure in Document Number (calc '3' vs extracted '9').	62.0% (HIGH)	SECONDARY INSPECT
Preset 4: DOB Mismatch	Dmitri Volkov (RUS • N77109234)	Visual zone DOB '01.01.1999' contradicts encoded MRZ DOB '1985-03-14'.	62.0% (HIGH)	SECONDARY INSPECT
Preset 5: Interpol Red Notice	Viktor Korshikov (RUS • L898902C3)	Active Interpol Red Notice hit: Transnational Identity Fraud & Document Forgery.	92.0% (CRITICAL)	DETAIN SUBJECT
Custom IRL Profile: Live Passenger Test	Custom Passenger (Real User Ingestion)	Live webcam snapshot verified against uploaded document; ICAO check digits audited.	Dynamic (0–95%)	Automated Routing

7. Cryptographic Audit Ledger & Compliance Certification

Every screening transaction automatically generates a cryptographically verifiable SHA-256 audit record. The record binds officer badge, checkpoint ID, timestamp, extracted traveler metadata, risk score, and enforcement action into an immutable digital certificate admissible in border enforcement court proceedings.

Sample Cryptographic Audit Certificate:

- **Audit Certificate ID:** BCP-1772904120-9FB1E559
- **Inspection Station:** DELHI-IGI-T3-GATE-14 • **Officer ID:** OFFICER-7419 (B. DAS)
- **SHA-256 Digest:** 9fb1e5591c02bbf3a71b9d4e8c1874220dcfae56214589d71e2c9081e81249fa
- **Timestamp:** 2026-09-06T00:18:29+05:30 • **Security Standard:** ICAO DOC 9303 / ISO 19794-5

8. Deployment Verification & GitHub Repository

The complete source code, documentation, backend modules, and frontend UI have been pushed to GitHub:

GitHub Repository: <https://github.com/Biswarup-das23/AI-Based-Fake-Identity-Document-Screening-System>

Both the FastAPI backend (Port 8000) and the Vite frontend (Port 5173) are fully operational and ready for production demonstrations.